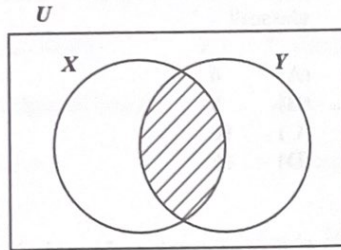


1. The number 3.14063 written correct to 3 decimal places is
- (A) 3.140
(B) 3.141
(C) 3.146
(D) 3.150
2. What percentage of 40 is 8?
- (A) 5
(B) 20
(C) 32
(D) 150
3. In standard notation, 208.06 is written as
- (A) 2.0806×10^{-2}
(B) 20.806×10^1
(C) 2.0806×10^2
(D) 0.20806×10^3
4. What is the LEAST number of apples that can be shared **equally** among either 6, 10 or 15 children?
- (A) 24
(B) 30
(C) 60
(D) 90
5. If \$350 is shared in the ratio 2:5, the LARGER share is
- (A) \$ 50
(B) \$ 70
(C) \$100
(D) \$250
6. There are 40 students in a class. Girls make up 60% of the class. 25% of the girls wear glasses. How many girls in the class wear glasses?
- (A) 6
(B) 8
(C) 10
(D) 15
7. If the universal set $U = \{1, 2, 3, 4, 5, 6\}$ and $H = \{3, 4, 6\}$, then $H' =$
- (A) $\{1, 3, 5\}$
(B) $\{1, 2, 5\}$
(C) $\{2, 4, 6\}$
(D) $\{3, 4, 6\}$
8. Which of the following statements describes the set of integers from -3 to 6?
- (A) $\{x : -3 > x > 6, x \in \mathbb{Z}\}$
(B) $\{x : -3 \geq x \geq 6, x \in \mathbb{Z}\}$
(C) $\{x : -3 \leq x \leq 6, x \in \mathbb{Z}\}$
(D) $\{x : -3 < x < 6, x \in \mathbb{Z}\}$
9. If X and Y are 2 finite sets such that $n(X) = 9$, $n(Y) = 10$ and $n(X \cup Y) = 15$, then $n(X \cap Y)$ is
- (A) 4
(B) 6
(C) 10
(D) 19

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Item 10 refers to the following Venn diagram.



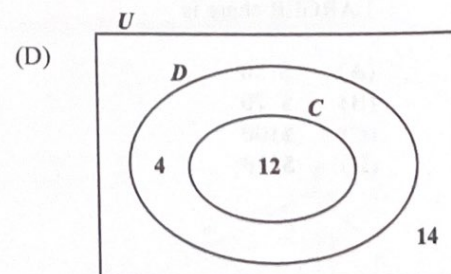
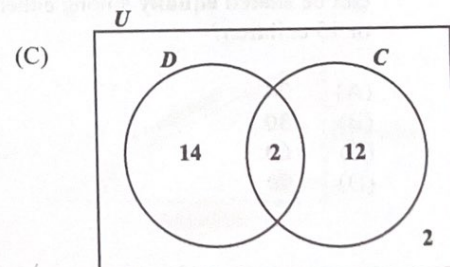
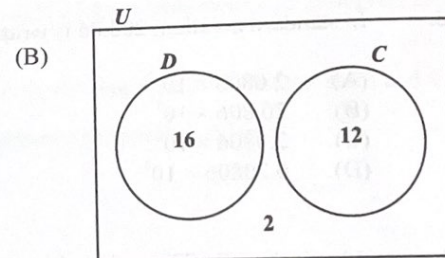
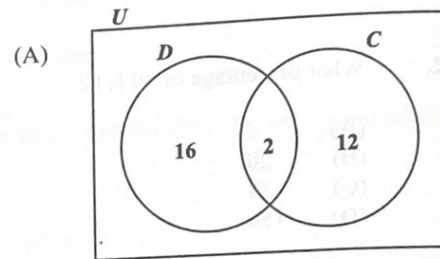
10. In the Venn diagram, X represents the set of factors of 12. Y represents the set of factors of 9. The shaded region represents the set of all factors of

- (A) 3
- (B) 6
- (C) 21
- (D) 108

11. Which of the following pairs of sets is an example of disjoint sets?

- (A) $X = \{\text{whole numbers}\}$ and $Y = \{\text{rational numbers}\}$
- (B) $G = \{\text{multiples of five}\}$ and $H = \{\text{multiples of ten}\}$
- (C) $P = \{\text{multiples of 2}\}$ and $Q = \{\text{multiples of 3}\}$
- (D) $E = \{\text{even numbers}\}$ and $F = \{\text{odd numbers}\}$

12. The 30 students in Teacher May's class have either a dog (D) or a cat (C), or none of the two. ALL the students who have a cat also have a dog. If $n(C) = 12$ and $n(D) = 16$, which of the following Venn diagrams correctly represents this information?



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13. Excluding sales tax, how much will be saved when a video which costs \$12.00 is sold at a 20% discount?
- (A) \$0.24
(B) \$1.20
(C) \$2.40
(D) \$3.60
14. If EC\$2.50 is equivalent to US\$1.00, then EC\$20.00 in US dollars is
- (A) \$ 5.00
(B) \$ 8.00
(C) \$50.00
(D) \$80.00
15. The cash price of a television set is \$350. When bought on hire purchase, a deposit of \$35 is required, followed by 12 monthly payments of \$30. How much is saved by paying cash?
- (A) \$10
(B) \$25
(C) \$40
(D) \$45
16. A company employs 20 persons as gardeners and clerks. The **mean** daily wage of ALL the employees is \$22.40. If EACH of the 12 gardeners is paid \$26.00 per day, then the daily amount paid to each of the 8 clerks is
- (A) \$ 3.60
(B) \$12.20
(C) \$17.00
(D) \$48.40
17. A man's annual salary is \$45 000. His tax free allowances total \$13 000. He has to pay a tax of 20% on his taxable income.
- The tax payable is
- (A) \$ 2 600
(B) \$ 6 400
(C) \$ 9 000
(D) \$11 600
18. A salesman sells a car for \$10 500. If he is paid a commission of 5% for the first \$10 000 and 8% on the remainder, then the TOTAL commission he receives is
- (A) \$525
(B) \$530
(C) \$540
(D) \$565
19. A man's basic wage for a 40-hour week is \$160.00. He is paid \$5.00 per hour for overtime. If he works $6\frac{1}{2}$ hours overtime in a certain week, then his wage for that week is
- (A) \$165.00
(B) \$166.50
(C) \$171.50
(D) \$192.50
20. A calculator which is marked at \$120 is sold for cash at a 30% discount. How much change would Susan receive if she pays for the calculator with a \$100 bill?
- (A) \$16
(B) \$20
(C) \$28
(D) \$36

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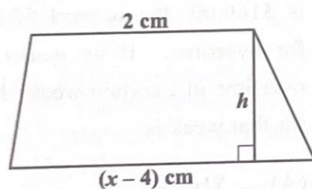
21. Meghan normally saves \$ x each month but in August she saved \$4 less than **twice** her usual amount. In August, she saved

(A) \$4 x
 (B) \$6 x
 (C) \$2 $x - 4$
 (D) \$2($x - 4$)

22. If $m * n = \sqrt{m^3 - n^2}$, then $5 * (5 * 2) =$

(A) 2
 (B) $\sqrt{11}$
 (C) $\sqrt{34}$
 (D) 11

Item 23 refers to the following diagram of a trapezium.



23. The trapezium with height, h , has an area of $x^2 \text{ cm}^2$. The height of the trapezium expressed in terms of x is

(A) $h = \frac{2x^2}{x-2}$
 (B) $h = \frac{x-2}{2x^2}$
 (C) $h = \frac{x^2}{2(x-4)}$
 (D) $h = \frac{x^2}{(x-4)(x+2)}$

24. $3 - (x - 2) =$

(A) $1 - x$
 (B) $x + 5$
 (C) $5 - x$
 (D) $3x + 2$

25. $5^{n+1} \times 5^{n+2}$ is the same as

(A) 5^{2n}
 (B) 5^{2n+3}
 (C) $5^{3(2n)}$
 (D) 2×5^{2n}

26. If $5(y - 2) = 3(y + 4)$, then y is

(A) -1
 (B) 3
 (C) 7
 (D) 11

Item 27 refers to the following vectors, $\mathbf{p}$ and $\mathbf{q}$.

$$\mathbf{p} = \begin{bmatrix} 3 \\ 7 \end{bmatrix} \quad \mathbf{q} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$$

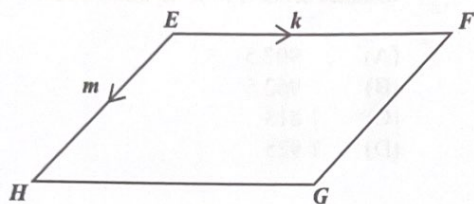
27. The vector $\mathbf{p} - 2\mathbf{q}$ is represented by

(A) $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$
 (B) $\begin{bmatrix} -7 \\ 3 \end{bmatrix}$
 (C) $\begin{bmatrix} 7 \\ -3 \end{bmatrix}$
 (D) $\begin{bmatrix} -1 \\ -3 \end{bmatrix}$

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Item 28 refers to the following diagram of a parallelogram, in which EF is parallel to HG , EH is parallel to FG , $\overrightarrow{EF} = k$ and $\overrightarrow{EH} = m$.



28. $\overrightarrow{EG}$ expressed in terms of k and m is

- (A) $k + m$
- (B) $k - m$
- (C) $m - k$
- (D) $-m - k$

29. If $5 \begin{pmatrix} x \\ y \end{pmatrix} = 4 \begin{pmatrix} 10 \\ 20 \end{pmatrix}$, then the values of x and y are

- (A) $x = 4, y = 5$
- (B) $x = 8, y = 16$
- (C) $x = 2.5, y = 4$
- (D) $x = 10, y = 20$

Item 30 refers to the following matrix, A .

$$A = \begin{pmatrix} 2 & 1 \\ 7 & 4 \end{pmatrix}$$

30. The inverse of A , A^{-1} , is given by

- (A) $\frac{1}{17} \begin{pmatrix} 4 & -1 \\ -7 & 2 \end{pmatrix}$
- (B) $\frac{1}{17} \begin{pmatrix} -4 & 1 \\ 7 & -2 \end{pmatrix}$
- (C) $\begin{pmatrix} -4 & 1 \\ 7 & -2 \end{pmatrix}$
- (D) $\begin{pmatrix} 4 & -1 \\ -7 & 2 \end{pmatrix}$

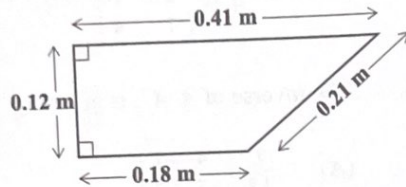
31. If it took a speedboat 9 hours to travel a distance of 1 080 km, what was its average speed, in kmh^{-1} ?

- (A) 12
- (B) 102
- (C) 120
- (D) 1 200

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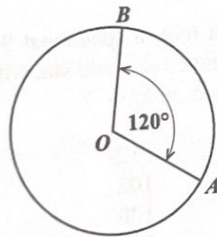
Item 32 refers to the following diagram of a trapezium.



32. The perimeter of the trapezium, in cm, is

- (A) 0.092
- (B) 9.2
- (C) 92
- (D) 920

Item 33 refers to the following circle, with centre O .



33. If the circumference of the circle is 15 cm, then the length of the minor arc AB , in cm, is

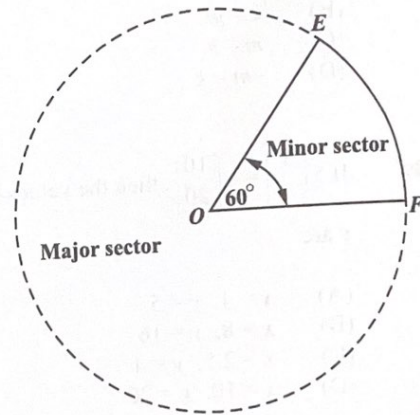
- (A) $\frac{360}{120} \times 15$
- (B) $\frac{120}{360} \times 15$
- (C) $\frac{360}{360 - 120} \times 15$
- (D) $\frac{360 - 120}{360} \times 15$

34.

An aircraft leaves A at 16:00 hours and arrives at B at 19:30 hours, travelling at an average speed of 550 kilometres per hour. A and B are in the same time zone. The distance from A to B , in kilometres, is

- (A) 907.5
- (B) 962.5
- (C) 1 815
- (D) 1 925

Item 35 refers to the following diagram which shows a circle, with centre O , and the major and minor sectors indicated.



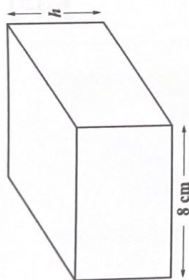
35. If the area of the major sector is 515 cm^2 , then the area of the circle, in cm^2 , is

- (A) 103
- (B) 412
- (C) 575
- (D) 618

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Item 36 refers to the following diagram which shows a cuboid.



Item 37. The area of a triangle is 60 cm^2 and its height is 10 cm . What is the length of the base of the triangle, in cm ?

- (A) 6
- (B) 12
- (C) 13
- (D) 17

Item 38. A square has the same perimeter as a rectangle whose sides are 9 cm and 15 cm in length. What is the length, in cm , of the side of the square?

- (A) 9
- (B) 12
- (C) 12.5
- (D) 50

Item 39. The volume of the cuboid is 320 cm^3 and the height is $h \text{ cm}$. If the length of the square base of the cuboid is 8 cm , what is the value of h ?

- (A) 5 cm
- (B) 16 cm
- (C) 32 cm
- (D) 64 cm

Items 39 and 40 refer to the following frequency distribution which shows the average mass, in kg , of a group of children in a school.

Mass (kg)	21–30	31–40	41–50	51–60	61–70
Frequency	29	47	24	39	11

Item 39. The lower limit of the median class is

- (A) 31
- (B) 41
- (C) 51
- (D) 61

Item 40. How many children have a mass of at MOST 50 kg ?

- (A) 24
- (B) 50
- (C) 76
- (D) 100



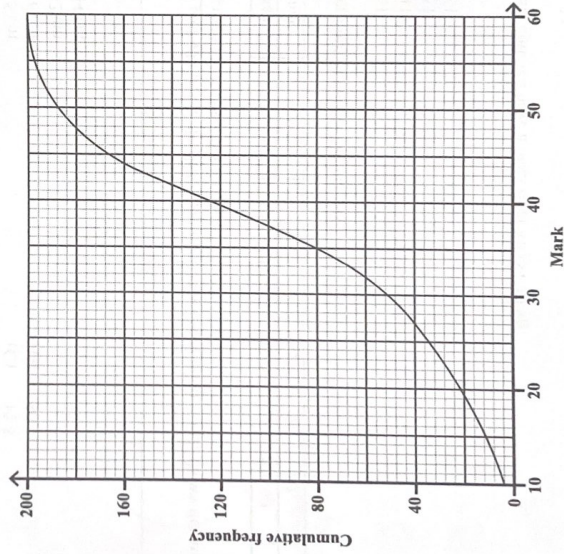
Item 41 refers to the following table which shows the frequency distribution of scores obtained by students on a test.

Score	2	3	5	6	8	11
Number of Students	8	4	6	3	12	2

41. The modal score is

- (A) 8
- (B) 9
- (C) 10
- (D) 12

Item 42 refers to the following diagram of a cumulative frequency curve which shows the marks obtained by 200 students on a test.



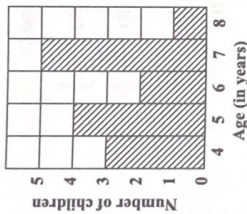
42. The lower quartile of the marks scored by the 200 students is

- (A) 30.0
- (B) 35.0
- (C) 37.5
- (D) 45.0

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Item 43 refers to the following bar chart which shows the ages of children who took part in a survey.



43. A child is selected at random. What is the probability that the child is MORE than 4 years old?

- (A) $\frac{1}{5}$
- (B) $\frac{1}{15}$
- (C) $\frac{13}{15}$
- (D) $\frac{4}{5}$

44. Six hundred students sit an examination. The probability of a randomly selected student passing the examination is $\frac{4}{5}$.

How many students are expected to fail?

- (A) 80
- (B) 100
- (C) 120
- (D) 480

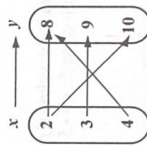
45. Which of the following functions represents the equation of a straight line?

- (A) $y = \frac{4}{x}$
- (B) $y = x^2 - 4$
- (C) $y = 2x + 3$
- (D) $y = x^2 + 2x - 5$

46. The point where a linear function crosses the vertical axis is

- (A) always negative
- (B) always positive
- (C) the x-intercept
- (D) the y-intercept

Item 47 refers to the following arrow diagram.



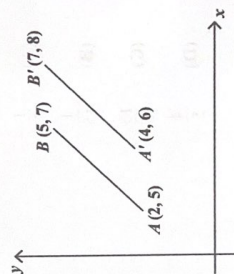
47. The arrow diagram above describes the relation

- (A) y is less than x
- (B) x is a factor of y
- (C) y is a factor of x
- (D) x is a multiple of y



48. What is the y-intercept of the straight line $2y = -3x - 8$?
 (A) 2
 (B) 3
 (C) $-\frac{3}{2}$
 (D) -4
49. If $h(x) = \frac{3x-2}{5}$, then $h(4) =$
 (A) -4
 (B) $-\frac{14}{5}$
 (C) 2
 (D) $\frac{14}{5}$
52. A line L is perpendicular to the line $y = \frac{2}{5}x - 8$. What is the gradient of the line L ?
 (A) $-\frac{5}{2}$
 (B) $-\frac{2}{5}$
 (C) $\frac{2}{5}$
 (D) 2

Item 53 refers to the following diagram which shows a translation.



50. What is the value of $f^{-1}(1)$, if the function f is given as $f: x \rightarrow 5 - 2x$?
 (A) 1
 (B) 2
 (C) 3
 (D) 7

53. In the diagram, the translation by which AB is mapped onto $A'B'$ is represented by

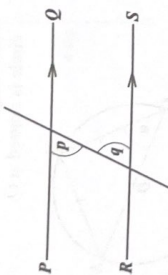
- (A) $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$
 (B) $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$
 (C) $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$
 (D) $\begin{pmatrix} 5 \\ 3 \end{pmatrix}$

51. The values of x at the points where $y = 4x - x^2$ intersects the x-axis are

- (A) $x = 0$ and $x = 2$
 (B) $x = 2$ and $x = 4$
 (C) $x = 0$ and $x = 4$
 (D) $x = 0$ and $x = -4$



Item 54 refers to the following transversal diagram in which PQ and RS are parallel.



54. Which of the following statements BEST describes the relation between p and q ?

- (A) $p = q$
- (B) $p < q$
- (C) $p + q = 90^\circ$
- (D) $p + q = 180^\circ$

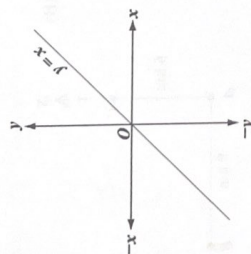
55. A ship sailed 8 km due east from A to B . It then sailed 6 km due north to C . Which of the following diagrams BEST represents the path of the ship?

- (A)
- (B)
- (C)
- (D)

56. In a polygon, Angle $A = x^\circ$ and Angle A and Angle B are supplementary. What is the size of Angle B ?

- (A) 30°
- (B) 60°
- (C) $90^\circ - x$
- (D) $180^\circ - x$

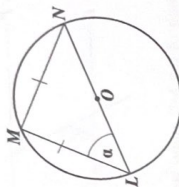
Item 57 refers to the following diagram which shows the graph of the line $y = x$.



57. If the line $y = x$ is rotated anti-clockwise about O through 90° , then its image is

- (A) $y = -x$
- (B) $x = y$
- (C) $x = 0$
- (D) $y = 0$

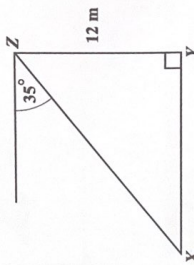
Item 58 refers to the following diagram which shows a triangle within a circle. In the triangle, LM and MN are equal. The circle is centred at O .



58. What is the size of Angle α ?

- (A) 25°
- (B) 45°
- (C) 50°
- (D) 90°

Item 59 refers to the following diagram which shows that the angle of depression of a point X from Z is 35° .

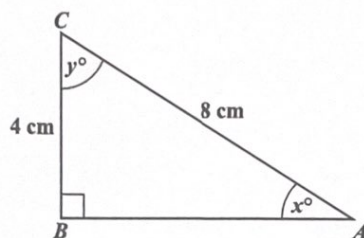


59. If Y is 12 metres from Z , then the base, XY , in metres, is

- (A) $12 \cos 35^\circ$
- (B) $12 \sin 35^\circ$
- (C) $12 \tan 35^\circ$
- (D) $12 \tan 55^\circ$



Item 60 refers to the following right-angled triangle.



60. Which trigonometric ratio is equal to $\frac{4}{8}$?

- (A) $\sin x$
- (B) $\tan y$
- (C) $\cos x$
- (D) $\tan x$

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

